



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	DIGITAL SIGNAL PROCESSING				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Fundamental Mathematics Concepts: sequences and Series, Integration, Matrices. Fourier Theorem 2. Basic idea of transducers.				
Course Summary	This course outlines the fundamentals of signal processing by digital means.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the characteristics of discrete-time signals and systems.	U	C	Quizzes, homework assignments, exams

CO2	Apply the Z-transform to analyse discrete-time signals and systems.	Ap	P	Problem-solving exercises, projects
CO3	Analyse the frequency content of discrete-time signals using the Z-transform.	An	C	Homework assignments, exams
CO4	Design discrete-time filters for specific signal processing tasks.	C	M	Laboratory assignments, projects
CO5	Implement signal processing algorithms using digital signal processing tools.	Ap	P	Projects, simulations
CO6	Interpret and evaluate the performance of signal processing systems.	E	C	Case studies, presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	INTRODUCTION		8	15
	1	Signals, Systems. and Signal Processing	1	
	2	Classification of Signals	1	
	3	The Concept of Frequency in Continuous-Time and Discrete-Time Signals	2	
	4	Analog-to-Digital and Digital-to-Analog Conversion	3	

	5	Digital-to-Analog Conversion	1	
	Sections 1.1, 1.2, 1.3 (1.3.1 and 1.3.2 only), 1.4 (1.4.1 to 1.4.6) of Book 1			
II	DISCRETE-TIME SIGNALS AND SYSTEMS		14	20
	6	Discrete-Time Signals	2	
	7	Discrete-Time Systems	2	
	8	Analysis of Discrete-Time Linear Time-Invariant Systems	4	
	9	Discrete-Time Systems Described by Difference Equations	3	
	10	Correlation of Discrete-Time Signals	3	
	Sections 2.1, 2.2, 2.3, 2.4, 2.6 (2.6.1 and 2.6.2 only) of Book 1.			
III	THE Z-TRANSFORM		13	15
	11	The z Transform	3	
	12	Properties of the z-Transform	3	
	13	Rational z-Transforms	3	
	14	Inversion of the z-Transform	4	
	Sections 3.1, 3.2, 3.3, 3.4 of Book 1			
IV	FREQUENCY ANALYSIS OF DISCRETE-TIME SIGNALS		13	20
	15	The Fourier Series for Discrete-Time Periodic Signals	2	
	16	Power Density Spectrum of Periodic Signals.	1	
	17	The Fourier Transform of Discrete-Time Aperiodic Signals.	1	
	18	Relationship of the Fourier Transform to the z-Transform	1	
	19	The Cepstrum	1	
	20	Properties of the Fourier Transform for Discrete-Time	2	

		Signals		
	21	Frequency Domain Sampling: The Discrete Fourier Transform	3	
	22	Properties of the DFT	2	
	Sections 4.2(4.2.1, 4.2.2,4.2.3,4.2.6,4.2.7), 4.4, 7.1(7.1.1, 7.1.2, 7.1.3), 7.2(7.2.1, 7.2.2) of Book 1			
V	OPEN ENDED MODULE: FILTERS		12	
	1	Various filters like lowpass filter, highpass filter, bandpass filter, Bandpass filter, Notch filter, comp filter etc Digital resonators Digital sinusoidal oscillators	12	
	Sections from References: 5.4 from Book 1			
Books and References:				
1. Digital Signal Processing: Principles, Algorithms, and Applications. John G. Proakis, Dimitris G. Manolakis Fourth edition. (Book 1)				
2. Digital Signal Processing Oppenheim, Alan V (Book 2)				
3. Digital Signal Processing Ramesh Babu, P (Book 3)				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	2	2	1	3	3	3	2	3	1	2	3	3
CO 2	3	2	2	2	3	3	3	2	3	1	2	3	3
CO 3	3	3	3	2	3	2	3	2	3	1	2	3	3
CO 4	3	2	2	3	3	3	3	2	3	1	2	3	3
CO 5	3	2	2	3	3	3	3	2	3	1	2	3	3
CO 6	3	2	3	2	3	3	3	2	3	1	2	3	3

Correlation Levels:

Level	Correlation
0	Nil

1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	